

Computer Language and Programming (Spring semester, 2026)

Final Project: Smart GPA Planner & Academic Management System

Group members: 蔡日玲 (2025080090), 陈奕婷 (2025080094), 杜黄灵 (2025080098),

范梅英 (2025080099) , 郑京珠 (2025080129)

1. Overview

This project is a GPA planner and academic management tool written in C for Tsinghua University students, designed for a four-year program with eight semesters. It helps students track their grades and graduation progress by using features like GPA prediction, semester performance reviews, grade distribution reports etc. You can add, edit, delete, search, and sort course records, and the system automatically calculates both your overall and major GPAs. All your academic data is saved automatically, making it easy to access whenever you need it. Overall, the system helps students manage their grades, monitor their progress, and stay on track to graduate. Examples of functionality can be seen below in the program functionalities part.

2. Language and IDE

The development of this project is carried out using the C programming language. C is commonly used for building structured and reliable applications. In this project, the language was used to develop core functions such as GPA calculation, course management, academic record

analysis etc. It is done by making use of functions, arrays, and structures which was taught throughout the course. The integrated development environment (IDE) used for this program development is Visual Studio Code, which is developed by Microsoft. Below are the standard libraries that were used in the code.

stdio.h

The stdio.h library is used for input, output, and file handling in the program. It allows the program to display menus and messages on the screen, receive information from the user, and save or load student data files. For example, printf() is used to display the main menu, scanf() is used to receive user choices and course information, and fopen() is used to open files when saving or loading student records.

stdlib.h

The stdlib.h library is used for general utility functions. It helps the program convert text entered by the user into numbers, sort course records, and run system commands. For example, atof() converts a grade entered as text into a floating-point number, atoi() converts text into an integer, and qsort() is used when sorting courses by grade, semester, or name.

string.h

The string.h library is used to work with text data such as student names, course names,

student IDs, and majors. It helps the program copy, compare, search, and manage strings. For example, `strcpy()` is used to copy course names, `strcmp()` is used to compare text, and `strcspn()` is used to remove the newline character after the user enters a name.

ctype.h

The `ctype.h` library is used for checking and processing characters. It helps the program identify whether the user entered a letter grade or a numerical grade and allows text comparisons without being affected by uppercase or lowercase letters. For example, `isalpha()` checks whether an input such as "A" or "B+" is a letter grade, while `tolower()` is used when comparing course names and grades in a case-insensitive manner.

math.h

The `math.h` library is used for mathematical operations involving decimal numbers. In this program, it is mainly used to check whether an entered grade matches one of the valid grade values. For example, the `fabs()` function calculates the absolute difference between two floating-point numbers when validating grades such as 4.0, 3.6, or 3.3.

3. Program functionalities

The table below summarises all 14 menu features together with their purpose:

Menu	Feature	Description
------	---------	-------------

1	Student Profile	Create / update name, student ID, major, required & major credits
2	Course Management	Add, edit, delete, sort (by semester / grade / name), and search courses; supports P/F course, major flag
3	GPA Calculator	Compute and display cumulative GPA and major-only GPA; P/F courses are excluded from all GPA calculations
4	Semester Trend Analysis	Per-semester GPA table with UP / STABLE / DOWN trend indicator
5	Advanced Transcript	Full course listing grouped by semester with grade letters
6	Credit Tracker	Total and major credits completed vs. requirements; progress %
7	Course Statistics	Highest, lowest, and average grade across all courses (P/F courses excluded)
8	Grade Distribution	Count of A / B / C / D / F courses with ASCII bar chart (P/F courses excluded)
9	Graduation Audit	Automated check of credit, major-credit, and GPA requirements
10	What-If Analyzer	Shows required avg. GPA in remaining credits to hit five target GPAs
11	Final GPA Predictor	Calculates needed per-credit GPA to reach a user-specified target
12	Export to CSV	Writes full transcript as CSV file to ../other/<filename>.csv
13	Import from CSV	Parses a CSV and bulk-adds valid courses; skips duplicates / invalid
14	Academic Dashboard	Summary panel: GPA, credits, progress %, and Honours classification

Input / Output Formats

Feature	Input	Output
Add Course	Name, credits (1-6), semester (1-8), P/F flag, grade (skipped if P/F), major flag	Confirmation message; data saved to .dat file
GPA Calculator	Reads stored courses (except for P/F courses)	Cumulative GPA and Major GPA to 3 decimal places
What-If Analyzer	None (reads stored courses)	Table of 5 target GPAs vs. required average; feasibility flag
Final GPA Predictor	Target GPA (float)	Required average grade; advice string; max achievable if impossible
Export to CSV	Output filename (string)	../other/<name>.csv ; with header rows and one course per line
Import from CSV	CSV filepath (string)	Count of imported courses; [SKIP] messages for rejected rows

Grade input accepts two equivalent formats

⑩ Numeric point value: 4.0, 3.6, 3.3, 3.0, 2.6, 2.3, 2.0, 1.6, 1.3, or 0.0

⑩ Letter grade: A+, A, A-, B+, B, B-, C+, C, C-, D+, D, or F

Invalid entries are rejected with a re-prompt loop; duplicates are blocked by a case-insensitive course-name check.

4. Examples and demonstration

Example 1 : Adding a Course (Option 1)

User selects menu option 2 and 1 (Add Course) and enters:

⑩ Course name: Economics Principles 1

⑩ Credits: 3

⑩ Semester: 1

⑩ Grade: 4.0 (accepted as either number or letter, stored as 4.00 (A+))

⑩ Major course: 1 (Yes)

Output: [OK] Course added successfully! — data is immediately saved to student_data.dat.

```
C:\Users\AD\OneDrive\Docum x + v
TSINGHUA UNIVERSITY GPA PLANNER
=====
1. Student Profile Management
2. Course Management
3. GPA Calculator
4. Semester Trend Analysis
5. Advanced Transcript
6. Credit Tracker
7. Course Statistics
8. Grade Distribution
9. Graduation Audit
10. What-If Scenario Analyzer
11. Final GPA Predictor
12. Export to Excel (CSV)
13. Import from CSV
14. Academic Dashboard
0. Save & Exit
=====
Your choice: 2

--- COURSE MANAGEMENT ---
1. Add Course
2. View All Courses
3. Edit Course
4. Delete Course
5. Sort by Semester
6. Sort by Grade
7. Sort by Name
8. Search by Name
9. Search by Semester
0. Back
Choice: 1

=== ADD NEW COURSE ===
Course name: Economics Principles 1
Credits (1-6): 3
Semester (1-8): 1
Is this a P/F course? (1=Yes, 0=No): 0
Grade (4.0,3.6,3.3,3.0,2.6,2.3,2.0,1.6,1.3,0.0 or A+,A-,B+,B-,C+,C-,D+,D,F): 4.0
Is this a major course? (1=Yes, 0=No): 1

[OK] Course added successfully!
[OK] Data saved successfully!
```

Adding a duplicate course

If the student attempts to add a course with the same name as one already stored (case-insensitive), the program blocks the entry immediately:

Course name: Economics Principles 1

[ERROR] Course 'Economics Principles 1' already exists!

The check is performed by `course_exists()`, which compares the new name against all stored course names using `my_strcasecmp()` — so "Economics Principles 1", "ECONOMICS PRINCIPLES 1", and "economics principles 1" are all treated as the same course. The record is not added and the user is returned to the menu.

```
--- COURSE MANAGEMENT ---
1. Add Course
2. View All Courses
3. Edit Course
4. Delete Course
5. Sort by Semester
6. Sort by Grade
7. Sort by Name
8. Search by Name
9. Search by Semester
0. Back
Choice: 1

=== ADD NEW COURSE ===
Course name: Economics Principles 1
[ERROR] Course 'Economics Principles 1' already exists!

--- COURSE MANAGEMENT ---
1. Add Course
2. View All Courses
3. Edit Course
4. Delete Course
5. Sort by Semester
6. Sort by Grade
7. Sort by Name
8. Search by Name
9. Search by Semester
0. Back
Choice:
```

Example 2 : GPA Calculator (option 3)

After entering several courses the student selects option 1 to back to the main menu and option 3.

The program prints:

Cumulative GPA: 3.815

Chinese 1	2 credits	Grade: 3.60 (B+)
Economics Principles 1	3 credits	Grade: 4.00 (A+) [Major]

[Semester 2]

Economics Principles 2	3 credits	Grade: 4.00 (A+) [Major]
Calculus 2	3 credits	Grade: 4.00 (A+)
Accounting Principles	3 credits	Grade: 3.60 (B+) [Major]
PE 2	1 credits	Grade: 4.00 (A+)
English 2	2 credits	Grade: 4.00 (A+)
Info Management Intro	3 credits	Grade: 4.00 (A+) [Major]
Computer Language and Programming	3 credits	Grade: 4.00 (A+)
Freshman Seminar	2 credits	Grade: 4.00 (A+)
Chemistry and Society	2 credits	Grade: 0.00 [P/F]

=====

Cumulative GPA: 3.815

Major GPA: 3.900

=====

Courses are sorted by semester before display. Each row shows course name, credits, numeric grade, letter grade, and [Major], [P/F] tags for major and pass/fail courses. Cumulative and major GPA are printed at the end.

```

C:\Users\AD\OneDrive\Docum X + v
7. Course Statistics
8. Grade Distribution
9. Graduation Audit
10. What-If Scenario Analyzer
11. Final GPA Predictor
12. Export to Excel (CSV)
13. Import from CSV
14. Academic Dashboard
0. Save & Exit
=====
Your choice: 5
[OK] Courses sorted by semester!

=====
ADVANCED TRANSCRIPT
=====
Student: DO HOANG LINH (2025080098)
Major: SEM
=====

[Semester 1]
Calculus 1          3 credits  Grade: 3.60 (B+)
Linear Algebra      3 credits  Grade: 3.30 (B)
PE 1                1 credits  Grade: 3.00 (B-)
English 1           2 credits  Grade: 4.00 (A+)
Chinese 1           2 credits  Grade: 3.60 (B+)
Economics Principles 1 3 credits  Grade: 4.00 (A+) [Major]

[Semester 2]
Economics Principles 2 3 credits  Grade: 4.00 (A+) [Major]
Calculus 2           3 credits  Grade: 4.00 (A+)
Accounting Principles 3 credits  Grade: 3.60 (B+) [Major]
PE 2                 1 credits  Grade: 4.00 (A+)
English 2            2 credits  Grade: 4.00 (A+)
Info Management Intro 3 credits  Grade: 4.00 (A+) [Major]
Computer Language and Programming 3 credits  Grade: 4.00 (A+)
Freshman Seminar     2 credits  Grade: 4.00 (A+)
Chemistry and Society 2 credits  Grade: 0.00 (F) [P/F]

=====
Cumulative GPA: 3.815
Major GPA: 3.900
=====

```

Example 4 : What-If Analyzer (option 10)

With 36 credits completed, including 2 P/F (GPA 3.815) and 119 remaining, the output shows:

Target GPA	Required Avg in Remaining	Feasible?
3.00	2.77 (F)	YES
3.20	3.02 (F)	YES
3.50	3.41 (F)	YES
3.70	3.67 (F)	YES
4.00	4.05 (IMPOSSIBLE!)	NO

```
=====
TSINGHUA UNIVERSITY GPA PLANNER
=====
1. Student Profile Management
2. Course Management
3. GPA Calculator
4. Semester Trend Analysis
5. Advanced Transcript
6. Credit Tracker
7. Course Statistics
8. Grade Distribution
9. Graduation Audit
10. What-If Scenario Analyzer
11. Final GPA Predictor
12. Export to Excel (CSV)
13. Import from CSV
14. Academic Dashboard
0. Save & Exit
=====
Your choice: 10

=====
WHAT-IF SCENARIO ANALYZER
=====
Current GPA: 3.815
Current credits completed: 36 (incl. 2 P/F)
Remaining credits needed: 117

Scenario Analysis:
-----
Target GPA   Required Avg in Remaining   Feasible?
-----
3.00         2.76 (F)                   YES
3.20         3.02 (F)                   YES
3.50         3.41 (F)                   YES
3.70         3.67 (F)                   YES
4.00         4.05 (IMPOSSIBLE!)         NO
=====
```

Example 5 : Academic Dashboard (option 14)

The dashboard consolidates all key figures in one screen:

```
=====

TSINGHUA UNIVERSITY DASHBOARD

=====

Student: DO HOANG LINH

ID: 2025080098

Major: SEM

-----

Cumulative GPA: 3.815

Major GPA:      3.900

Credits: 36 / 153 (23.5%)

Major Credits: 12/50

-----

CLASSIFICATION (for reference only): First Class Honours (Excellent)

=====
```

```
=====
TSINGHUA UNIVERSITY GPA PLANNER
=====
1. Student Profile Management
2. Course Management
3. GPA Calculator
4. Semester Trend Analysis
5. Advanced Transcript
6. Credit Tracker
7. Course Statistics
8. Grade Distribution
9. Graduation Audit
10. What-If Scenario Analyzer
11. Final GPA Predictor
12. Export to Excel (CSV)
13. Import from CSV
14. Academic Dashboard
0. Save & Exit
=====
Your choice: 14

=====
TSINGHUA UNIVERSITY DASHBOARD
=====
Student: DO HOANG LINH
ID: 2025080098
Major: SEM
=====
Cumulative GPA: 3.815
Major GPA: 3.900
Credits: 36 / 153 (23.5%)
Major Credits: 12 / 50
=====
CLASSIFICATION (for reference only): First Class Honours (Excellent)
=====
```

Example 6 : Adding a Pass/Fail Course (Option 2 → 1)

The student adds a Chemistry and Society course graded on a Pass/Fail basis. They select option

2 then 1 (Add Course) and enter:

- ⑩ Course name: Chemistry and Society
- ⑩ Credits: 2
- ⑩ Semester: 2
- ⑩ P/F course: 1 (Yes) — grade prompt is skipped automatically
- ⑩ Major course: 0 (No)

Output: [OK] Course added successfully!

Note: This P/F course counts toward credits but NOT toward GPA.

The course now appears in transcripts as

Chemistry and Society 2 credits Grade: 0.00 [P/F]

and its 2 credit is added to the total credit count. However, it does not change the cumulative or major GPA since P/F courses are excluded from all GPA calculations.

```
TSINGHUA UNIVERSITY GPA PLANNER
=====
1. Student Profile Management
2. Course Management
3. GPA Calculator
4. Semester Trend Analysis
5. Advanced Transcript
6. Credit Tracker
7. Course Statistics
8. Grade Distribution
9. Graduation Audit
10. What-If Scenario Analyzer
11. Final GPA Predictor
12. Export to Excel (CSV)
13. Import from CSV
14. Academic Dashboard
0. Save & Exit
=====
Your choice: 2

--- COURSE MANAGEMENT ---
1. Add Course
2. View All Courses
3. Edit Course
4. Delete Course
5. Sort by Semester
6. Sort by Grade
7. Sort by Name
8. Search by Name
9. Search by Semester
0. Back
Choice: 1

=== ADD NEW COURSE ===
Course name: Chemistry and Society
Credits (1-6): 2
Semester (1-8): 2
Is this a P/F course? (1=Yes, 0=No): 1
Is this a major course? (1=Yes, 0=No): 0

[OK] Course added successfully!
Note: This P/F course counts toward credits but NOT toward GPA.
[OK] Data saved successfully!
```

5. Program highlights

One of the main features of the system is GPA calculation and prediction. The program can automatically calculate both cumulative GPA and major GPA based on the courses entered by the user. It also includes GPA prediction and what-if analysis functions, which help students estimate the grades they need in future semesters.

Another important feature is credit tracking and graduation auditing. Students can track completed credits, check their graduation progress, and see whether they have met graduation

requirements. This helps users keep track of their academic status more easily.

The system also includes data management functions. The system supports saving data to files and exporting or importing records through CSV files, which can be opened and edited in Microsoft Excel. This makes it easier for students to manage their academic information and keep their records for future use.

6. Originality and external library

This project was designed and implemented as a GPA planning and academic management system for university students. The idea came from our own experience as students, as we often need to calculate GPA, track academic progress, and check graduation requirements throughout our studies. We wanted to develop a simple and convenient system that could bring these functions together in one application. The project combines multiple functions, including course management, GPA prediction, transcript generation, credit tracking, and graduation auditing, making it easier for students to manage their academic information in one place. The overall program structure, data management system, GPA calculation functions, graduation tracking features, and CSV import/export functions were designed and implemented by our team.

There are no external or third-party libraries used in this project. The program was developed using only standard C libraries, including `stdio.h`, `stdlib.h`, `string.h`, `ctype.h`, and `math.h`. As a result, no additional dependencies are required, and the program can be compiled and executed in a

standard C programming environment. This makes the program easier to compile, use, and maintain.